Ratios and Rates

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KEY WORDS ratio *(ōwehenga)* · simplify *(whakamāmā)* · equivalent *(ōrite)* · share · part · rate *(pāpātanga)* · per

A Simplifying ratios ●○○

Remember: Divide every part of the ratio by the same number, as you do when you simplify a fraction.

1. Write each ratio in its simplest form.

a) 4 : 6 = _____ **b)** 10 : 15 = _____ **c)** 9 : 3 = _____ **d)** 12 : 20 = _____

e) 24 : 16 = _____ **f)** 35 : 14 = _____ **g)** 18 : 27 = _____ **h)** 50 : 125 = _____

i) 6 : 9 : 12 = _____ **j)** 20 : 10 : 30 = _____ **k)** 45 : 60 = _____ **l)** 28 : 42 = _____

B Equivalent ratios ●●○

Remember: Find what the known part has been multiplied or divided by, and do the same to the other part.

2. Find the missing number.

a) $2 : 3 = 8 : \square$ **b)** $1 : 5 = 4 : \square$ **c)** $3 : 4 = \square : 12$ **d)** $5 : 2 = 20 : \square$

e) $7 : 3 = \square : 9$ **f)** $4 : 9 = 20 : \square$ **g)** $6 : 5 = \square : 25$ **h)** $3 : 8 = 15 : \square$

C Sharing in a ratio ●●○

Remember: Add the parts of the ratio. Divide the amount by that total to find one part, then multiply.

3. Share each amount.

a) Share 20 in the ratio 1 : 3.

b) Share 35 in the ratio 2 : 3.

c) Share 48 in the ratio 5 : 3.

d) Share \$60 in the ratio 1 : 4.

e) Share \$120 in the ratio 3 : 5.

f) Share 90 kg in the ratio 2 : 7.

g) Share 72 in the ratio 1 : 2 : 3.

h) Share \$150 in the ratio 2 : 3 : 5.

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D Finding another part ●●●

⚠ **Watch out:** Here you are given one part, not the total. Divide the amount by its own share of the ratio to find one part.

4. Answer each question.

a) The ratio of boys to girls in a club is 3 : 4. There are 12 boys. How many girls are there?

b) Paint is mixed from blue and yellow in the ratio 2 : 5. A painter uses 6 litres of blue. How much yellow is needed?

c) A recipe uses flour and sugar in the ratio 5 : 2. Ana uses 300 g of sugar. How much flour does she need?

d) The ratio of cats to dogs at a shelter is 4 : 7. There are 28 dogs. How many cats are there?

E Rates ●●○

Remember: A rate compares two different quantities. "Per" means "for each one": 60 km in 2 hours is 30 km per hour.

5. Find each rate.

a) \$15 for 5 kg of apples: cost per kg

b) 180 km in 3 hours: km per hour

c) \$96 for 8 hours of work: pay per hour

d) 450 words in 9 minutes: words per minute

e) 24 litres in 6 minutes: litres per minute

f) \$5.40 for 6 pens: cost per pen

6. Use the rate to answer each question.

a) A tap fills a bath at 12 litres per minute. How much water runs in 8 minutes?

b) Tama earns \$23.50 per hour. How much does he earn in 6 hours?

c) A car travels at 85 km per hour. How far does it travel in 3 hours?

★ Extension

7. Two numbers are in the ratio 3 : 5. The larger number is 16 more than the smaller number. Find the larger number.

8. A 400 g box of cereal costs \$3.60. Find the cost per 100 g.

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Ratios and Rates

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS ratio *(ōwehenga)* · simplify *(whakamāmā)* · equivalent *(ōrite)* · share · part · rate *(pāpātanga)* · per

A Simplifying ratios ●○○

Remember: Divide every part of the ratio by the same number, as you do when you simplify a fraction.

1. Write each ratio in its simplest form.

a) 6 : 8 = _____ **b)** 9 : 15 = _____ **c)** 12 : 4 = _____ **d)** 15 : 25 = _____

e) 27 : 18 = _____ **f)** 21 : 35 = _____ **g)** 16 : 24 = _____ **h)** 75 : 100 = _____

i) 4 : 6 : 10 = _____ **j)** 30 : 15 : 45 = _____ **k)** 36 : 60 = _____ **l)** 32 : 48 = _____

B Equivalent ratios ●●○

Remember: Find what the known part has been multiplied or divided by, and do the same to the other part.

2. Find the missing number.

a) $2 : 5 = 6 : \square$ **b)** $1 : 4 = 5 : \square$ **c)** $3 : 5 = \square : 20$ **d)** $7 : 2 = 21 : \square$

e) $5 : 3 = \square : 15$ **f)** $4 : 7 = 20 : \square$ **g)** $8 : 5 = \square : 25$ **h)** $3 : 10 = 12 : \square$

C Sharing in a ratio ●●○

Remember: Add the parts of the ratio. Divide the amount by that total to find one part, then multiply.

3. Share each amount.

a) Share 24 in the ratio 1 : 3.

b) Share 40 in the ratio 3 : 5.

c) Share 54 in the ratio 4 : 5.

d) Share \$75 in the ratio 1 : 4.

e) Share \$160 in the ratio 3 : 5.

f) Share 63 kg in the ratio 2 : 7.

g) Share 84 in the ratio 1 : 2 : 3.

h) Share \$200 in the ratio 2 : 3 : 5.

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D Finding another part ●●●

⚠ **Watch out:** Here you are given one part, not the total. Divide the amount by its own share of the ratio to find one part.

4. Answer each question.

a) The ratio of boys to girls in a club is 2 : 5. There are 14 boys. How many girls are there?

b) Paint is mixed from red and white in the ratio 3 : 4. A painter uses 9 litres of red. How much white is needed?

c) A recipe uses flour and butter in the ratio 7 : 2. Ana uses 150 g of butter. How much flour does she need?

d) The ratio of cats to dogs at a shelter is 3 : 8. There are 40 dogs. How many cats are there?

E Rates ●●○

Remember: A rate compares two different quantities. "Per" means "for each one": 60 km in 2 hours is 30 km per hour.

5. Find each rate.

a) \$18 for 6 kg of apples: cost per kg

b) 240 km in 4 hours: km per hour

c) \$91 for 7 hours of work: pay per hour

d) 400 words in 8 minutes: words per minute

e) 35 litres in 7 minutes: litres per minute

f) \$7.20 for 8 pens: cost per pen

6. Use the rate to answer each question.

a) A tap fills a bath at 14 litres per minute. How much water runs in 7 minutes?

b) Tama earns \$24.50 per hour. How much does he earn in 8 hours?

c) A car travels at 95 km per hour. How far does it travel in 3 hours?

★ Extension

7. Two numbers are in the ratio 4 : 7. The larger number is 18 more than the smaller number. Find the larger number.

8. A 500 g box of cereal costs \$4.25. Find the cost per 100 g.

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